Multi-stage events

1 An archer makes two attempts to hit a target, and the probability that he hits the target on any one attempt is $\frac{1}{3}$. Find the probability that the archer will miss the target on both attempts.

2 Christa enters a competition in which she guesses the 3-digit code (from 000 to 999) which cracks open a vault containing one million dollars. If the 3-digit number to open the vault is randomly generated by a computer, find the probability that it is:

a An odd number.

b An even number (including 000).

c A number greater than 123.

d A number divisible by 10.

e A number less than 321.

3 A fair die is rolled twice. Find the probability of:

a Rolling a 6 and a 1 in any order.

b Rolling a 6 and then a 1.

4 A computer generates two random numbers between 1 and 100 (inclusive). Using the product rule, find the probability that the two numbers are:

a The same.

b Different.

c 89 and 83 in that order.

d 89 and 83 in any order.

e 14 and 36 in any order.

f Not 16 and 95.

5 A computer generates a two-digit number by randomly selecting the first digit from $1, 3, 5, 7$ and 9, and randomly selecting the second digit from $0, 2, 4, 6$ and 8. Using the product rule, find the probability that number generated is:

a 94 **b** Even **c** Odd **d** Less than 32

e Greater than 74

6 In a game of Monopoly, rolling a double means rolling the same number on both dice. When you roll a double this allows you to have another turn. Find the probability that Sarah rolls:

a A double 1.

b A double 5.

c Any double.

d Two doubles in a row.

7 Two dice are rolled, and the numbers appearing on the uppermost faces are added. Find:

a $P(\text{a total less than } 6)$

b $P(\text{a total greater than or equal to } 10)$

c $P(\text{a total of at least } 7)$

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8 James is in the middle of a game of Monopoly, which involves players rolling two dice at a time. James is desperate to reach the Go! square and collect $\$200$. Calculate the probability that he lands exactly on the Go! square if it is:

a 2 squares ahead of his current position.

b 3 squares ahead of his current position

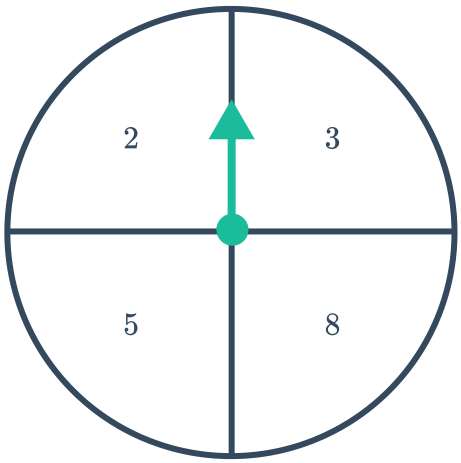
c 4 squares ahead of his current position

d 5 squares ahead of his current position

Tables

9 The following spinner is spun and a normal six-sided die is rolled. The product of their respective results is recorded.

	2	3	5	8
1				
2				
3				
4		12		
5				
6				48



a Complete the table above to represent all possible outcomes.

b State the total number of possible outcomes.

c Find the probability of:

i An odd product.

ii A 5 on the dice and scoring an even product.

iii A 3 on the spinner or scoring a product which is a multiple of 4.

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10 Two dice are rolled, and the combination of numbers rolled on the dice is recorded.

a Complete the table of outcomes:

	1	2	3	4	5	6
1	1,1	1,2				
2	2,1					
3						
4						
5						
6						

b Find the following probabilities for the two numbers rolled:

i $P(1 \text{ and } 4)$

ii $P(1 \text{ then } 4)$

iii $P(\text{difference} = 4)$

iv $P(\text{product} = 12)$

v $P(\text{difference} \leq 2)$

vi $P(\text{difference} = 3)$

vii $P(\text{product} = 20)$

viii $P(\text{difference} \leq 1)$

c The numbers appearing on the uppermost faces are added. State whether the following are true.

i A sum greater than 7 and a sum less than 7 are equally likely.

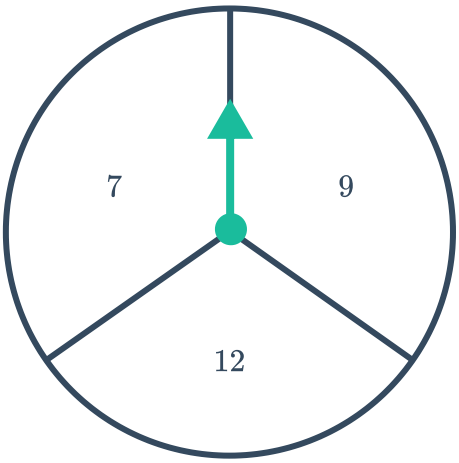
ii A sum greater than 7 is more likely than a sum less than 7.

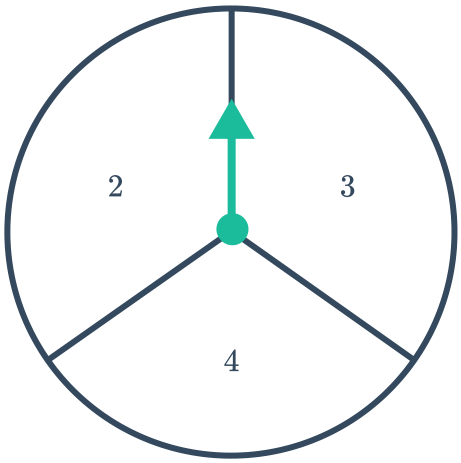
iii A sum of 5 or 9 is more likely than a sum of 4 or 10.

iv An even sum is more likely than an odd sum.

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11 The following two spinners are spun and the sum of their results are added:





a Complete the given table to represent all possible outcomes:

b State the total number of possible outcomes.

c Find the probability that:

i The first spinner lands on an even number and the sum is even.

ii The first spinner lands on a prime number and the sum is odd.

iii The sum is a multiple of 4.

	7	9	12
2	9		
3			13
4			

12 Two standard six-sided dice are rolled at the same time.

a Write out all the possible outcomes in a table.

b Find the probability of tossing two heads.

c Find the probability of tossing at least one head.

13 A standard six-sided die is rolled and a coin is tossed at the same time.

a Write out all the possible outcomes in a table.

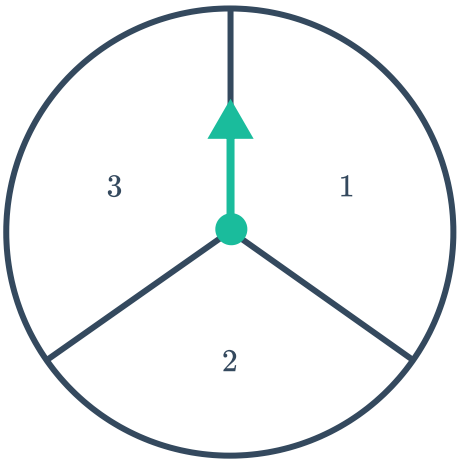
b Find the probability of tossing a heads and rolling an even number.

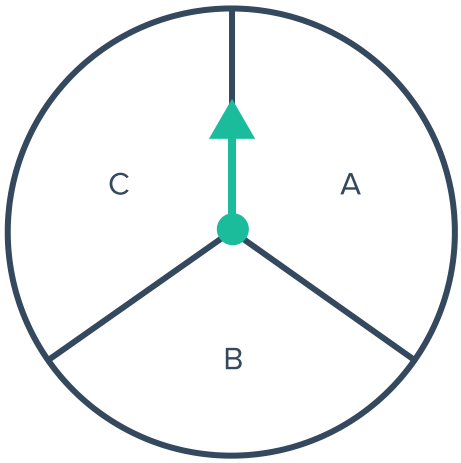
c Find the probability of tossing a heads or rolling an even number.

d Find the probability of tossing a tails and rolling a number greater than 2.

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14 The following two spinners are spun and the result of each spin is recorded:





a Complete the given table to represent all possible combinations:

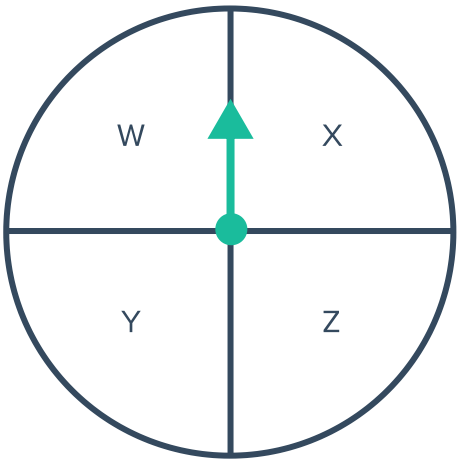
b State the total number of possible outcomes.

c Find the probability that the spinner lands on a consonant and an even number.

Spinner	A	B	C
1	1,A	,	,
2	,	,	2,C
3	,	,	,

d Find the probability that the spinner lands on a vowel or a prime number.

15 The following spinner is spun and a normal six-sided die is rolled. The result of each is recorded:



	W	X	Y	Z
1	1,W	,	,	,
2	,	,	,	2,Z
3	,	,	,	,
4	,	,	,	,
5	,	5,X	,	,
6	,	,	,	,

a Complete the table above to represent all possible combinations.

b State the total number of possible outcomes.

c Find the probability that the spinner lands on X and the dice rolls a prime number.

d Find the probability that the spinner lands on W and the dice rolls a factor of 6.

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e Find the probability that the spinner doesn't land on Z or the dice doesn't roll a multiple of 3.

Lists

16 Luke can't remember the order of his pin for his EFTPOS card, but he knows it contains the digits in 2853. Listed below are all the possible combinations of Luke's pin beginning with 2 and 8:

$$2853, 2385, 8523, 2835, 2358, 8532, 2583, 8253, 8325, 2538, 8235, 8352$$

a List all the other possible combinations of pins.

b State the total number of possible outcomes.

c Find the probability his pin starts with 5.

d Find the probability his pin has 5 followed immediately by 2.

e Find the probability his pin starts with 8, or ends with 3, or both.

17 Buzz can't remember the combination for his lock, but he knows it is a three digit number and contains the digits 6, 8 and 9.

a List all possible locker combinations that Buzz should try.

b State the total number of possible outcomes.

c If Buzz is correct that the combination includes 6, 8 and 9, find the probability that:

i The combination starts with 6.

ii The combination starts with 6 and ends with 8.

iii The combination starts with 6 or ends with 9.

18 A three-digit number is to be formed from the digits 4, 5 and 9, where the digits cannot be repeated.

a List all the possible numbers in the sample space.

b Find the probability that the number formed is:

i Odd

ii Even

iii Less than 900.

iv Divisible by 5.

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Tree diagrams

19 A coin is tossed twice.

- **a** Construct a tree diagram to identify the sample space of tossing a coin twice.
- **b** Use the tree diagram to find the probability of getting two tails.
- **c** Use the tree diagram to find the probability of getting at least one tail.

20 Ivan rolled a standard die and then tossed a coin.

- **a** Construct a tree diagram to identify the sample space of rolling a standard die and then tossing a coin.
- **b** List all the possible outcomes in the sample space.
- **c** Find the probability of the result including an odd number.
- **d** Find the probability that the result includes a number less than or equal to 5, and a tail.

21 Han has a special three-sided die with the numbers $1, 2$ and 3 on its sides. He rolls it and then tosses a coin.

- **a** Construct a tree diagram to list all possible outcomes.
- **b** Using the tree diagram, find the probability of a 1 and a tail.
- **c** Using the tree diagram, find the probability of an outcome with a head.
- **d** Using the tree diagram, find the probability of an outcome with an odd number.

22 Construct a tree diagram showing the following:

- **a** All possible outcomes of boys and girls that a couple with three children can possibly have.
- **b** All the ways the names of three candidates; Alvin, Sally and Peter, can be listed on a ballot paper.

23 Every morning Mae has toast for breakfast. Each day she either chooses honey or jam to spread on her toast, with equal chance of choosing either one.

- **a** Construct a tree diagram for three consecutive days of Mae's breakfast choices.
- **b** Find the probability that on the fourth day Mae chooses honey for her toast.
- **c** Find the probability that Mae chooses jam for her toast three days in a row.

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24 A die is rolled twice.

a Construct a tree diagram showing all the possible results of the given experiment.

b Use the tree diagram to find the probability of rolling:

i A double 5.

ii The same number twice.

iii Two different numbers.

iv Two odd numbers.

25 On the island of Timbuktoo the probability that a set of traffic lights shows red, yellow or green is equally likely. Christa is travelling down a road where there are two sets of traffic lights.

a Construct a tree diagram to indicate the possible pairs of traffic lights.

b Find the probability that both sets of traffic lights will be yellow.

26 Three fair coins are tossed.

a Construct the tree diagram for the experiment given.

b Find the probability of obtaining:

i At least one head.

ii TTH in this order.

iii THH in this order.

Probability trees

27 James owns four green jackets and three blue jackets. He selects one of the jackets at random for himself and then another jacket at random for his friend.

a Construct a probability tree of this situation with the correct probability on each branch.

b Find the probability that James selects a blue jacket for himself.

c Find the probability that both jackets James selects are green.

28 An archer has three arrows that each have a probability of $\frac{1}{5}$ of striking a target. If all three arrows are shot at a target:

a Construct a probability tree showing all the possible outcomes and probabilities.

b Find the probability that all three arrows will hit the target.

c Find the probability that at least one arrow will miss the target.

d Find the probability that at least one arrow will hit the target.

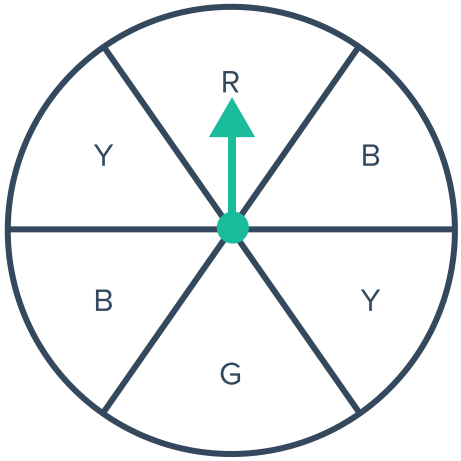
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29 Luke plays three tennis matches. In each match he has 60% chance of winning.

- **a** Construct a probability tree showing all his possible outcomes and probabilities in these three matches.
- **b** Find the probability that he will win all his matches.
- **c** Find the probability that he will lose all his matches.
- **d** Find the probability that he will win more matches than he loses.

30 A fair coin is tossed and then the following spinner is spun:

- **a** Construct a probability tree representing the situation.
- **b** Find the probability of getting a tail and then a yellow.
- **c** Find the probability of getting a tail, a yellow, or both.
- **d** Find the probability of getting a head and not getting a red.
- **e** Find the probability of not getting a head or a red.



31 One cube has 4 red faces and 2 blue faces, another cube has 3 red faces and 3 blue faces, and the final cube has 2 red faces and 4 blue faces. The three cubes are rolled like dice.

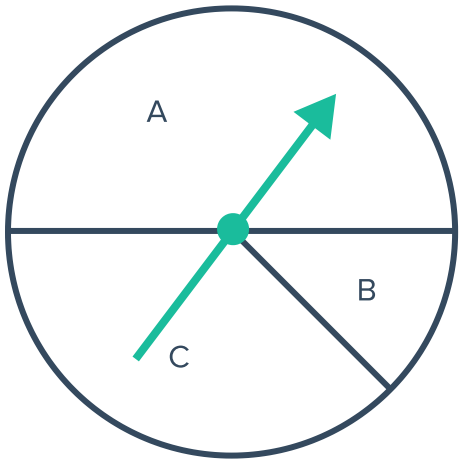
- **a** Construct a probability tree diagram that shows all possible outcomes and probabilities.
- **b** Find the probability that three red faces are rolled.
- **c** Find the probability that more red faces than blue faces are rolled.
- **d** Find the probability that only one cube rolls a blue face.

32 The proportion of scholarship recipients at a particular university is $\frac{7}{10}$. The number of students at the university is so large that even if a student is removed, we can say that the proportion of scholarship recipients remains the same. If three students are selected at random:

- **a** Construct a probability tree showing all the possible combinations of recipients and nonrecipients.
- **b** Find the probability that at least one of the students is a scholarship recipient.
- **c** Find the probability that at least one of the students is a nonrecipient.
- **d** Find the probability there is at least one recipient and one nonrecipient in the selection.

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33 A coin is tossed, then the spinner shown is spun and either lands on A, B or C.
Segment B is $\frac{1}{8}$ of the entire cirle.



a Construct a probability tree diagram showing all possible outcomes and probabilities.

b Find the probability of landing on tails and the spinner landing on A.

c Find the probability of landing on tails, or the spinner landing on A, or both.

34 For breakfast, Maria has something to eat and drinks a hot drink. She will either eat toast or cereal and will drink tea or Milo.

- The chance of Maria making toast is 0.7.
- The chance of Maria drinking Milo is 0.4.

a Construct a tree diagram illustrating all possible combinations of food and drink Maria can have for breakfast and their associated probabilities.

b Find the probability Maria drinks tea and eats toast.

c Find the probability Maria drinks tea or eats toast.

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